

Guorui Xiao

Mobile: 206-502-9268

Email: grxiao@cs.washington.edu

Website: xertxiao.github.io

RESEARCH INTEREST

Data management systems for large language models: making AI-powered data analysis cheaper, explainable, and trustworthy.

EDUCATION

- University of Washington** Seattle, WA, USA
• *Ph.D. Student, Computer Science; GPA: 3.93/4.0; Advisor: [Magdalena Balazinska](#) Expected: Sep. 2028*
- University of California, Los Angeles** Los Angeles, CA, USA
• *M.S., Computer Science; GPA: 4.0/4.0; Advisor: [Carlo Zaniolo](#) Mar. 2023*
- University of California, Los Angeles** Los Angeles, CA, USA
• *B.S., Computer Science; GPA: 3.77/4.0; Cum Laude Dec. 2020*

SELECTED PUBLICATIONS & MANUSCRIPTS

- [M1] **KathDB-FAO: Synthesized Query Plans in a Multimodal DBMS**
Guorui Xiao, Douglas Brown, Artur Borycki, and Magdalena Balazinska. In submission. ([arXiv](#))
- [P1] **KathDB: Explainable Multimodal Database Management System with Human-AI Collaboration**
Guorui Xiao, Enhao Zhang, Nicole Sullivan, Will Hansen, and Magdalena Balazinska. **CIDR** 2026.
- [P2] **CENTS: A Flexible and Cost-Effective Framework for LLM-Based Table Understanding**
Guorui Xiao, Dong He, Jin Wang, and Magdalena Balazinska. **VLDB** 2025, Vol. 18, No. 11, pp. 4574–4587.
- [P3] **Highly Efficient String Similarity Search and Join over Compressed Indexes**
Guorui Xiao, Jin Wang, Chunbin Lin, and Carlo Zaniolo. **ICDE** 2022, pp. 232–244.
- [P4] **RACOON: An LLM-based Framework for Retrieval-Augmented Column Type Annotation with a Knowledge Graph**
Lindsey Linxi Wei, Guorui Xiao, and Magdalena Balazinska. **NeurIPS** 2024 Table Representation Learning Workshop.

RESEARCH EXPERIENCE

- UW Database Group** University of Washington
• *Research Assistant Sep. 2023 – Present*
 - **KathDB [P1, M1]**: built an LLM-agent DBMS that answers natural-language queries over tables, text, and images, generating operators as code on the fly; cut execution cost by 58.8% on average vs. Claude Code with the same or better quality, while tracking provenance so users can ask why and how each answer was produced.
 - **CENTS [P2]**: designed a budget-aware framework that selects the table content an LLM needs, reducing input tokens by 95.1% on large tables while improving accuracy.

INDUSTRY EXPERIENCE

- Google LLC** Kirkland, WA, USA
• *SDE Intern Jun. 2026 – Sep. 2026*
 - Built non-linear proxy models, trained on a subset of LLM-labeled data, that answer BigQuery **AI.IF** / **AI.CLASSIFY** queries at scale, serving up to 2.2× more real customer traffic than the linear baseline.
- Apple, Inc.** Seattle, WA, USA
• *Research Intern Jun. 2025 – Sep. 2025*
 - Built a joint image–video training framework for a DiT video foundation model, so one model supports text-to-image, text-to-video, and image-to-video and improves over single-task baselines.

MENTORING

- Undergraduate Research Mentorship: [Lindsey Linxi Wei](#)** Seattle, WA, USA
• *Guided her project to a first-author NeurIPS 2024 workshop paper [P4] Jun. 2024 – Jun. 2025*

SKILLS

- **Languages:** Python, C/C++, Java, SQL, Bash, Datalog
- **Platforms:** DuckDB, Apache Spark, Apache Flink, LangGraph, Neo4j, Scikit-learn, AWS EC2